Reymart Salcedo Lagunero

University Assistant Postdoc – University of Graz, Austria

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Primary: Anton-Wildgans-Weg 15 / 42, Graz 8043 Austria
Secondary: Ave. de la Coruña 58 / 1°D, Las Rozas de Madrid 28231 Madrid, Spain
Birthdate: 12. August 1993 | Birthplace: Mayantoc, Tarlac Province 2304 Philippines

I am an **active member** of the **European Society for Mathematical and Theoretical Biology (ESMTB)**, **Mathematical Society of the Philippines (MSP)**, the **Southeast Asian Mathematical Society (SEAMS)**, and the **Philippine Society for Mathematical Biology (PSMB)**.

Education

30. Oct 2025 — **Dr.rer.nat. „mit Auszeichnung bestanden", University of Graz, Austria**
Dissertation: Lipolysis on Lipid Droplets: Mathematical Modelling, Analysis, and Numerical Simulations, **Supervisors:** Klemens Fellner Univ.-Prof. Dipl.-Ing. Dr.techn (Uni Graz) and Thomas Apel Univ.-Prof. Dr. rer. nat. habil. (Uni Bundeswehr Munich)
Completed in May 2025: Machine Learning and Big Data for Bioinformatics (Universidad de Granada Spain)
Visiting Researcher at the University of the Bundeswehr Munich: Feb 2022 - Jul 2022

21. Jun 2018 — **M.Sc., Master of Science (Mathematics), University of the Philippines, Philippines**
Master's Thesis: Well-posedness and Stability of Hyperbolic Systems of Partial Differential Equations with Delays, **Supervisor:** Gilbert Peralta Univ.-Assoc.Prof. Dr.rer.nat.
Visiting researcher: University of Graz Austria (Aug 2017 - Feb 2018)

28. Apr 2014 — **B.Sc., Bachelor of Science (Mathematics), University of the Philippines, Philippines**
Bachelor's Project: A study of the research paper entitled "Multivalued and Singlevalued Fixed Point Results in Partially Ordered Metric Spaces", **Supervisor:** Joel Addawe Univ.-Prof. PhD.
cum laude/with distinction

Professional Experience

Mar 2026 / Nov 2025 — **Department of Mathematics and Scientific Computing, University of Graz, Austria**
- 17.11.2025 - 15.03.2026, **University Assistant with Doctorate (Postdoc)**

Parameter Estimation of Kinetic Rates | Quasi-Steady-State-Approximations

Mar 2025 / Aug 2019 — **Department of Mathematics and Scientific Computing, University of Graz, Austria**
- 10.2024 - 03.2025, **University Assistant Predoc** and **Lecturer**, Master's Programme Practice Session: 24W Mathematical Modelling for Teachers in Training
- 01.2024 - 06.2024, **University Assistant Predoc** and **Lecturer**, Bachelor's Programme Practice Session: 24S Analysis 2 for Teachers in Training
- 10.2023 - 03.2024, **Lecturer**, Master's Programme Exercise Course: 23W Advanced Analysis
- 03.2023 - 09.2023, **Lecturer**, Master's Programme Exercise Course: 23S Calculus of Variations (with applications to Optimisation Theory)
- 10.2022 - 03.2023, **Lecturer**, Bachelor's Programme Exercise Course: 22W Computational Mathematics 1
- 07.2022 - 02.2023, **Visiting Researcher**, Institute for Mathematics & Computer-Based Simulation, University of the Bundeswehr Munich, Germany
- 10.2021 - 03.2022, **Lecturer**, Master's Programme Exercise Course: 21W Nonlinear Optimisation
- 08.2019 - 08.2023, **Academic Project Staff**, IGDK 1754 Munich - Graz: Optimization and Numerical Analysis of Partial Differential Equations with Nonsmooth Structures

Python | Matlab | FEniCS | GeoPDEs

Jul 2018 | **Actuarial Valuation Team, Willis Towers Watson Philippines, Bonifacio Global City**
May 2018 | **Actuarial Analyst**

- Conversion of Hong Kong client retirement plans from FORTRAN to high-level programming language
- Asset allocation reporting of Australia clients
- Valuation of retirement plans of company clients from Europe, Middle East and Africa (EMEA)
- Complete analysis and reconciliation of plan participant data
- Prepare valuation reports for consulting offices

Microsoft Office Excel and Word | High-level Programming

Jul 2019 | **Department of Mathematics and Computer Science, University of the Philippines, Baguio City**
Aug 2014

- 08.2018 - 07.2019, **Assistant Professor of Mathematics** (Full-time)
- 08.2016 - 07.2017, **Lecturer** (Part-time)
- 08.2014 - 07.2016, **Instructor** (Full-time)

Handled the following courses: Mathematical Methods in the Computational Sciences (BS Computer Science program), Analysis I (Calculus I), Analysis II (Calculus II), Analysis III (Calculus III), Mathematical Analysis (Advanced Analysis), Number Theory, Real Analysis, Basic Concepts in Mathematics, College Algebra and Plane Trigonometry, Plane Trigonometry, College Algebra, Elementary Statistics and Probability.

Matlab | Python | GeoGebra | R Language | C Language

Publications/Preprints/Scholarly Works

Published | Thomas Apel, Klemens Fellner, Volker Kempf, **Reymart Salcedo-Lagunero***, Philipp Zilk. *Lipolysis on Lipid Droplets: Mathematical Modelling and Numerical Discretisations*. Results Math Vol 81, Art Num 77 (2026) DOI:10.1007/s00025-026-02639-y

In preparation | **Reymart Salcedo Lagunero**, Klemens Fellner, and Cinzia Soresina. *Quasi-steady-state-approximations for lipolysis and transacylation on lipid droplets.*

In preparation | **Reymart Salcedo Lagunero** and Gilbert R. Peralta. *Rational and Exponential Stability of Hyperbolic Differential Equations with Distributed Delays.*

Dr.rer.nat. | **Reymart Salcedo Lagunero**. *Lipolysis on Lipid Droplets: Mathematical Modelling, Analysis, and Numerical Simulations.* Available Online at the University of Graz Library: `https://unipub.uni-graz.at/urn/urn:nbn:at:at-ubg:1-228640`

MSc. | **Reymart Salcedo Lagunero**. *Well-posedness and Stability of Hyperbolic System of Partial Differential Equations with Delays.* Access via UP Baguio Library at `https://tuklas.up.edu.ph/short/7745d8b2b`.

* Corresponding Author

Conferences Attended

Jun 2025 | **Oral Presentation**, Annual Convention of the Mathematical Society of the Philippines, Zamboanga City, Philippines. *„Isogeometric Analysis for Partial Differential Equations and Numerical Implementations in `GeoPDEs`"*

Jan 2025 | **Poster Presentation **Best Poster Award****, 16th Conference on Dynamical Systems Applied to Biology and the Natural Sciences, University of Naples Federico II, Naples, Italy. (see poster here) *„Lipolysis in Lipid Droplets: Modelling and Quasi-Steady-State-Approximations"*

May 2024 | **Oral Presentation**, Mathematical Society of the Philippines Annual Convention, University of the Philippines Tacloban, Philippines. *„Lipolysis in Lipid Droplets: Modelling, Quasi-Steady-State-Approximations, Numerical Simulations"*

Jun 2023	**Oral Presentation**, 50th Anniversary Convention of the Mathematical Society of the Philippines, Ateneo de Manila University, Quezon City, Philippines. *„Hydrolysis of Lipids: Mathematical Modelling and Numerical Simulations"*
Feb 2023	**Oral Presentation**, 14th Conference on Dynamical Systems Applied to Biology and Natural Sciences, Basque Center of Applied Mathematics, Bilbao, Spain. *„Lipid Hydrolysis: Modelling, Analysis, and Numerical Simulations"*
Sep 2022	**Oral Presentation**, 2nd Conference of Young Applied Mathematicians, Arenzano, Genoa, Italy. *„Numerical Simulations of a Reaction-Diffusion System for a Lipid Hydrolysis Test Model"*
May 2018	**Oral Presentation**, 45th Annual Convention of the Mathematical Society of the Philippines, Clark Freeport Zone, Pampanga, Philippines. *„Asymptotic stability of a system of hyperbolic partial differential equations damping and distributed delay"*
Jun 2016	**Participant**, 44th Annual Convention of the Mathematical Society of the Philippines, Cebu City, Philippines.

Research Schools/Workshops Attended

May 2024	**Centre International de Mathématiques Pures et Appliquées (CIMPA France)** Research School on Mathematics for Medicine and Health Sciences. Silpakorn University, Nakhon Pathom, Thailand.
Feb 2024	**Bio-PDE Days Vienna.** Technical University of Vienna, Austria.
Sep 2021	**ICMS - International Centre for Mathematical Sciences (Virtual)** MoDiS – Modelling Diffusive Systems: Theory and Biological Applications
Apr 2021	**Hausdorf School for Advanced Studies in Mathematics (Virtual)** Diffusive Systems: Pattern Formation, Bifurcations, and Biological Applications, Universität Bonn, Germany.
Dec 2020	**Erwin Schrödinger Institute (Virtual)** Mathematical Methods for the Study of Self-organization in the Biological Sciences, Vienna, Austria.
Jul 2017	**Department of Mathematics and Computer Science** Lecturer, Midyear School on Mathematics: Research Capability Building Series. University of the Philippines Baguio.
Jan 2017	**Centre International de Mathématiques Pures et Appliquées (CIMPA France)** Research School on the Introduction to the Mathematical Analysis of Differential Equations and Real-life Applications. National University of Laos, Vientiane Capital, Lao PDR.
Dec 2016	**South East Asian Mathematics Society (SEAMS)** School on Functional Analysis and Differential Equations. National University of Laos, Vientiane, Lao PDR.
Jun 2015	**Department of Mathematics and Computer Science** Summer School and Research Training: A Course on Graph Theory. University of the Philippines Baguio.

University Extension Services (University of the Philippines Baguio)

Jun 2019	**Program Development Associate**, ASEAN University Network Quality Assurance for BS Mathematics Program, University of the Philippines Baguio.
Feb 2019	**Facilitator**, Strategic Planning Seminar-Workshop of Don Mariano Marcos Memorial State University, Bacnotan, La Union.
Oct 2018	**Secretariat**, 2018 Mathematics Coaches Training, Mathematical Society of the Philippines, University of the Philippines Baguio.
2018-2019	**Faculty Adviser**, Amnesty International, University of the Philippines Baguio.

2018-2019	**Faculty Adviser**, UP Math-Physics Society, University of the Philippines Baguio.
2018-2019	**Chairperson**, College of Science Summer Institute in the Natural Sciences and Mathematics, University of the Philippines Baguio.
Apr 2016	**Co-Chairperson**, 22nd Summer Institute in the Natural Sciences and Mathematics, University of the Philippines Baguio.
2015-2016	**Faculty Adviser**, University Student Council of University of the Philippines Baguio. 2015-2016.
2013-2014	**President**, University of the Philippines Math-Physics Society, University of the Philippines Baguio.
2012-2013	**Chairperson**, External Affairs Committee of UP Math-Physics Society, University of the Philippines Baguio.

Languages, CEFRL

English	●●●●●●
German	●●○○○○
Spanish	●○○○○○
Filipino	●●●●●●

Competencies

Programming	Matlab, Python
Applications	GeoGebra, Wolfram Mathematica, MS Office (Word, Excel, PowerPoint), ParaView, LaTeX, Visual Studio Code
Operating System	Linux, Windows, macOS

Personal Interests

Outdoors:	Running, Hiking, Swimming, Volleyball, Badminton, Bowling, Archery
Indoors:	Cooking, Playing Guitar, Reading Books (non-academic)
Others:	Finished Camino de Santiago Portugués de la Costa (der Jakobsweg)
	Blood & Plasma Donor (Blood Galloner's Club - Philippine Red Cross, BioLife Plasmazentrum)
Favourite:	Travelling

References

Univ.-Prof. Dipl.-Ing. Dr.techn. KLEMENS FELLNER
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Univ.-Prof. Dr.rer.nat. habil. THOMAS APEL
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Univ.-Assist.Prof. Dr.rer.nat. MSc. GERVY MARIE MOLINA ANGELES
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